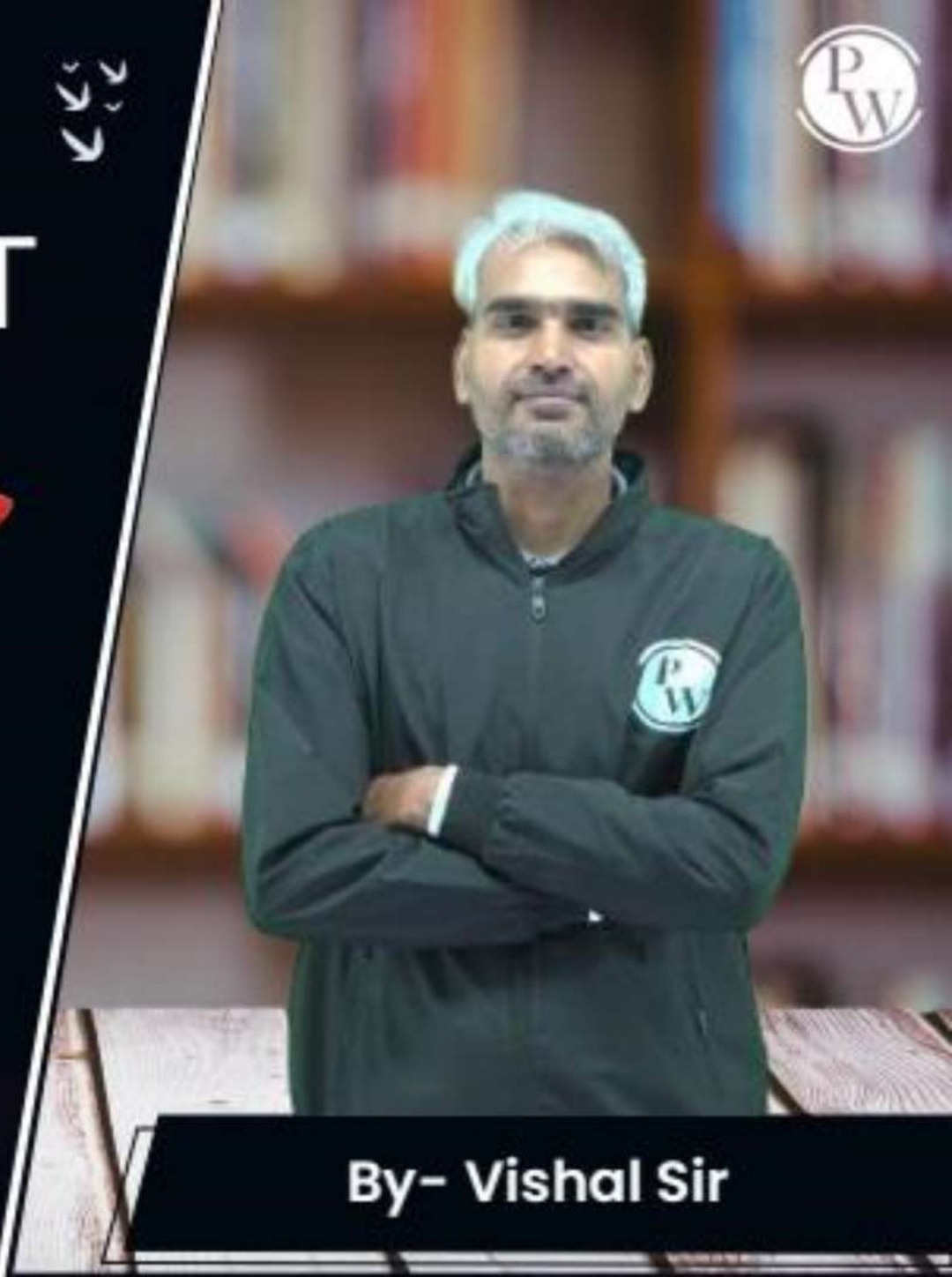
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 05



By- Vishal Sir

Recap of Previous Lecture



Topic Different types of keys in RDBMS

Slide

Topics to be Covered



- Topic: Closure of an attribute set
- Topic: Super key
- Topic: Candidate key
- Topic: Identification of candidate key w.r.t. FD set

Slide



Topic : Closure of an attribute set

Closure of an attribute set X (i.e., X^+) can be defined as set of all the attributes which can be functionally determined from attribute set X.



Topic : Algorithm to find Closure of an attribute set X

To find closure of an attribute set X (i.e., X^+) with respect to FD set F

```
X+ = X;
        while (X+ changes)
                { for each FD A → B in F
                        if A ⊆ X+ then X+ = X+ ∪ B
                }
```

Don't use this algorithm to Calculate Closure of a given attribute set

↓

it will Waste our time

#e.g. Assume a relation R (A,B,C,D) that has the following functional dependencies:

$$F = \{ A \to B,\ B \to C,\ C \to D \}$$

trivial

$(A)^+ = \{A, B, C, D\}$

$A \to B$

$B \to C$

$C \to D$

$(B)^+ = \{B, C, D\}$

$(C)^+ = \{C, D\}$

$(D)^+ = \{D\}$

$(AB)^+ = \{A, B, C, D\}$

$(AC)^+ = \{A, B, C, D\}$

$(AD)^+ = \{A, B, C, D\}$

$(BC)^+ = \{B, C, D\}$

$(BD)^+ = \{B, C, D\}$

$(CD)^+ = \{C, D\}$

$(ABC)^+ = \{A, B, C, D\}$

$(ABD)^+ = \{A, B, C, D\}$

$(ACD)^+ = \{A, B, C, D\}$

$(BCD)^+ = \{B, C, D\}$

$(ABCD)^+ = \{A, B, C, D\}$

#e.g. Assume a relation R (A,B,C,D) that has the following functional dependencies:

1. A→B
2. B→C
3. C→D

$\{A\}^+ = \{A\}$

first iteration

w.r.t. $A \to B$, $A \subseteq A^+$

$\therefore (A)^+ = A^+ \cup \{B\}$

$= \{A, B\}$

w.r.t. $B \to C$, $B \subseteq A^+$

$\therefore (A)^+ = A^+ \cup \{C\}$

$= \{A, B, C\}$

w.r.t. $C \to D$, $C \subseteq A^+$

$\therefore (A)^+ = A^+ \cup \{D\} = \{A, B, C, D\}$

Before first iteration

$(A)^+ = \{A\}$

after first iteration

$(A)^+ = \{A, B, C, D\}$

i.e. $(A)^+$ changes

∴ Repeat while loop

↓

We will keep doing this until the closure before starting the while loop and after completion of while loop are exactly same

#e.g. Consider the following FD set

$F = \{ AB \rightarrow CD, AF \rightarrow D, DE \rightarrow F, C \rightarrow G, F \rightarrow E, G \rightarrow A\}$

find the closure of following set of attributes.

(i) $\{C,F\} \Rightarrow \{C,F\}^+ = \{C, F, G, E, A, D\}$

(ii) $\{B,G\} \Rightarrow \{B,G\}^+ = \{B, G, A, C, D\}$

(iii) $\{A,F\} \Rightarrow \{A,F\}^+ = \{A, F, D, E\}$

(iv) $\{A,B\} \Rightarrow \{A,B\}^+ = \{A, B, C, D, G\}$



Topic : Super key

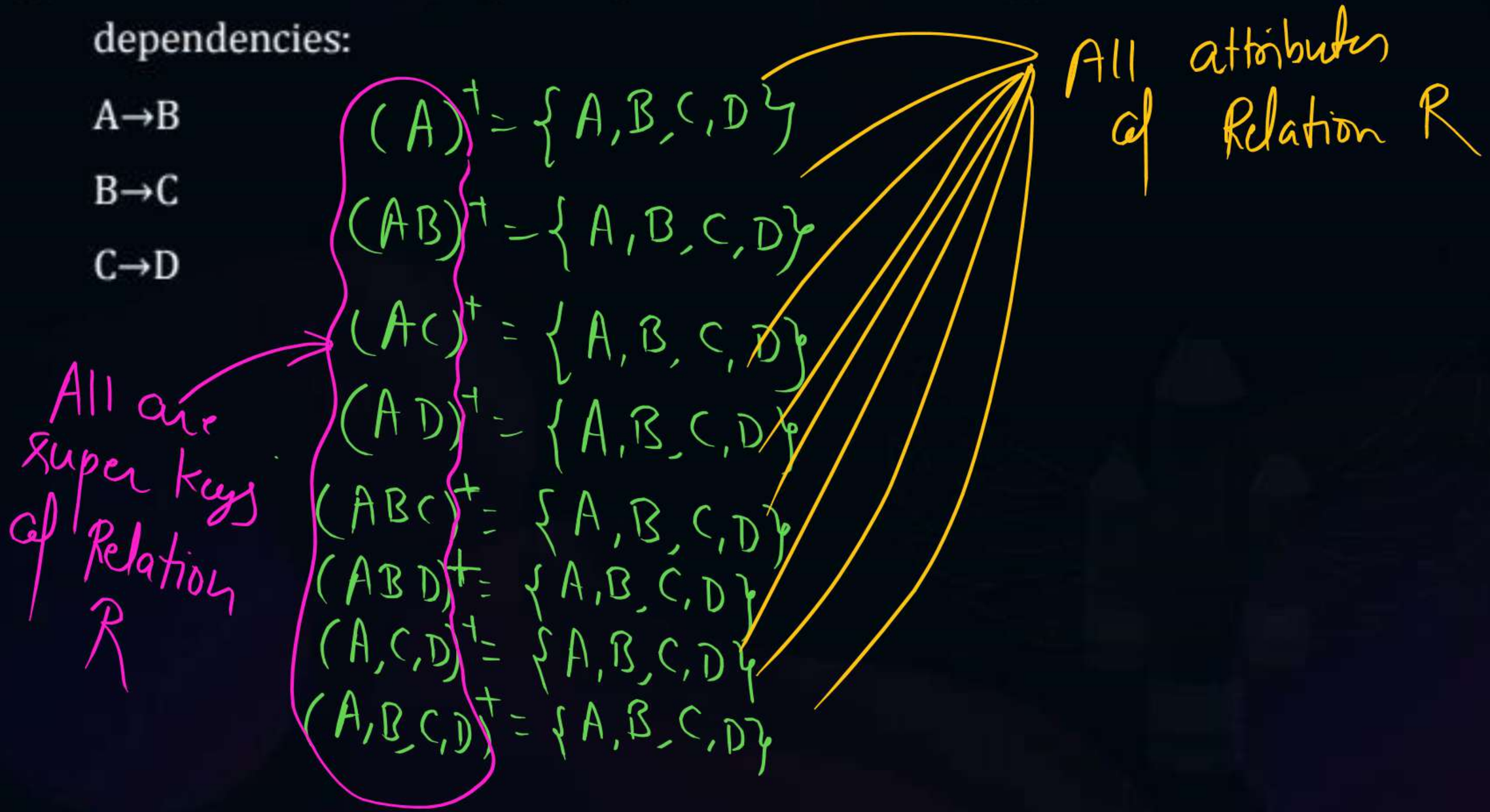
Let R be the relational schema, and let X be some attribute set over relation R. If X^+ determines all attributes of relation R, then X is called super key of relation R.

#e.g. Assume a relation R (A,B,C,D) that has the following functional dependencies:

A→B

B→C

C→D



* Every super key need not be minimal.

* Candidate keys are minimal keys



Topic : Candidate key (Minimal Super key)

X Can determine all attributes of R.

Let R be the relational schema, and let X be the super key of relation R.

If no proper subset of X is a super key then X is minimal super key

i.e., X is Candidate key

Candidate key

eg: let (A,B) is a super key of relation R

ie $\{A,B\}^+ = \{$ all attributes of R $\}$

Proper Subsets of $\{A,B\}$

$\{\} = \emptyset \Rightarrow \{\emptyset\}^+$ will always be $\emptyset$ it can never be a S.k. of any Rel^n

$\{A\} \Rightarrow \{A\}^+ = \{$ Not all attributes of R $\}$ i.e. $\{A\}$ is not a S.k

$\{B\} \Rightarrow \{B\}^+ = \{$ Not all attributes of R $\}$ i.e $\{B\}$ is not a S.k

i.e No Proper subset of $\{A,B\}$ is a S.k.
$\therefore \{A,B\}$ is a C.k.

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

$$F = \{AB \rightarrow C,\ B \rightarrow E,\ C \rightarrow D\}$$

Find the Candidate key of R.

If an attribute is not present in R.H.S. of any FD of FD set, then that attribute is called "essential attribute" of Candidate key. All "essential attributes" are always present in every Candidate key

* In the given question A & B are essential attributes.

$(AB)^+ = \{A, B, C, D, E\}$

all attributes of R

$\therefore \{AB\}$ is a S.K of R.

Proper Subsets of AB

$(A)^+ = \{A\}$, Not a S.K

$(B)^+ = \{B, E\}$, Not a S.K

$\Rightarrow \therefore \boxed{AB}$ is a C.K

For the given question,

AB is a candidate key

∴ Prime attributes = {A, B}

Prime Attributes A or B are not in the RHS of any FD of FD set

∴ There will be only one Candidate key of Relⁿ R

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

AB→C,

B→E,

C→D

E→A

Find the Candidate key of R.

Essential Attribute = {B}

$\{B\}^+ = \{B, E, A, C, D\}$

All attributes

∴ {B} is a Super key as well as a Candidate key {S.K. with single attribute}

$(AB)^+ = \{A, B, C, D, E\}$

all attributes of R

∴ AB is a S.K. of R.

Check whether AB is also a C.K or not

Proper subsets of {A,B}

$\{A\}^+ = \{A\}$ ∴ A is not a S.K

$\{B\}^+ = \{B, E, A, C, D\}$

all attributes ∴ B is a S.K

Proper subset of {A,B} is a S.K

∴ {A,B} can not be a Candidate key, because it is not Minimal

AB is just a Super key, but not a C.K

We found that B is a Super key of relation R,

B is a Super key with a single attribute,

∴ B is Candidate key of R

∴ Prime Attribute = {B}

Prime Attribute B is not in the RHS. of Any FD of FD set,

∴ Only one C.K for Rel^{n} R.

Note: Every super key with a single attribute is always a Candidate key



Topic : Note

{ $X \rightarrow$ Any Prime Attribute of Relation }

- If these exists any non-trivial FD $X \longrightarrow Y$ in relation R, s.t. Y is a prime attribute of relation R, then relation R will have more than one Candidate key.
- In corresponding candidate key we can replace prime attribute 'Y' by L.H.S. of FD $X \longrightarrow Y$, {i.e. replace Y by X}, to obtain a new Super key
  - then we can check for Candidate key

#Q. Assume a relation R (A, B, C, D) that has the following functional dependencies:

AB→CD,
D→A

Find all the Candidate keys of R.

B is the Essential attribute

$(AB)^+ = \{A, B, C, D\}$

- $(A)^+ = \{A\}$, Not a S.K
- $(B)^+ = \{B\}$, Not a S.K

No proper subset of {A, B} is a S.K

∴ AB is a C.K

Prime Attributes = {A, B}

in FD D → A

A is a prime attribute

Replace A by D

∴ We can replace A by D in C.K AB to obtain a new S.K.

DB is a S.K

$(D)^+ = \{D, A\}$
$(B)^+ = \{B\}$

No proper subset of DB is a S.K.

∴ DB is another C.K

hence Prime Attributes = {A, B, D}

New set of prime attributes = {A, D, B}

these exists an FD

$AB \rightarrow CD \equiv \underbrace{AB \rightarrow C \text{ \& } AB \rightarrow D}_{\text{Decomposition Rule w.r.t. FD}}$

in $AB \rightarrow D$

D is a prime attribute.

∴ we can replace D by AB in C.K. DB

So we get ABB ≡ AB which is already a CK

* No other replacement is possible

So Candidate Keys of Relation are AB and DB

H.W

#Q. Assume a relation R (M, N, O, P, Q) that has the following functional dependencies:

MNO → PQ and

P → MN

MNO → P
&
MNO → Q

Find the Candidate keys of R.

H.W.

#Q. Assume a relation R (A, B, C, D) that has the following functional dependencies:

$AB \rightarrow CD$

$C \rightarrow A$

$D \rightarrow B$

Find the Candidate keys of R.

H.W. Very good Concept to learn

#Q. Assume a relation R (A, B, C, D, E, H) that has the following functional dependencies:

$A \rightarrow B$

$BC \rightarrow D$

$E \rightarrow C$

$D \rightarrow A$

Find the Candidate keys of R.



2 mins Summary

Topic Closure of an attribute set

Topic Super key

Topic Candidate key

Slide

THANK - YOU